A NEURAL CLOSURE FOR A REDUCED MULTISCALE ATMOSPHERIC MODEL IN THE PRESENCE OF FAST OSCILATIONS AND RARE EVENTS

Romit Maulik*
School of Mechanical Engineering
Purdue University
585 Purdue Mall
rmaulik@purdue.edu

Santiago Parra
Department of Informatics
Università degli Studi di Verona
Verona, Italy
santiago.parramejia@studenti.univr.it

ABSTRACT

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1 Introduction

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*Use footnote for providing further information about author (webpage, alternative address)—*not* for acknowledging funding agencies.

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2 Formulation of the problem

The Lorenz-80 System results from a three mode Fourier truncation on the Primitive Equations (PE) of the atmosphere. The equations consist of a 9 dimensional system with variables $x, y, z \in \mathbb{R}^3$ and is given by:

$$a_i \frac{dx_i}{dt} = -v_0 a_i^2 x_i - c(a_i - a_k)x_j y_k + c(a_i - a_j)y_j x_k + a_i b_i x_j x_k - 2c^2 y_j y_k + a_i(y_i - z_i) \tag{1}$$

$$a_i \frac{dy_i}{dt} = -a_k b_k x_j y_k - a_j b_j y_j x_k + c(a_k - a_j)y_j y_k - a_i x_i - v_0 a_i^2 y_i \tag{2}$$

$$\frac{dz_i}{dt} = g_0 a_i x_i - b_k x_j (z_k - h_k) - b_j x_k (z_j - h_j) + c(z_k - h_k)y_j - c(z_j - h_j)y_k - k_0 a_i z_i + F_i \tag{3}$$

For each combination $(i, j, k) \in \{(1, 2, 3), (2, 3, 1), (3, 1, 2)\}$ where $x,\ y,\ z$ represent the fourier amplitudes of the velocity potential, the stream function, and the geopotential, respectively. In particular, the stream function represents the Rossby waves (which are large scale and change slowly with time), while the velocity potential is related to high frequency gravity waves. Thus, the system provides a simplified representation of multiscale weather dynamics.

As specified in Chekroun et al. [2024], these dynamics are dictated by the Rossby number Ro, which is defined by the square root of the constant force F_1. In particular, at small values of Ro, Rossby waves dominate over the gravity waves, in which case there is a frequency gap and it can be shown that the variables are driven by a smooth slow manifold which solves the closure problem for a variety of parameter regimes (see BE manifold in Chekroun et al. [2024], Chekroun et al. [2017]). However, as the Rossby number surpasses a critical limit Ro^*, this frequency gap disappears, and fast oscillations emerge in both slow and fast variables due to the exchange of frequencies Chekroun et al. [2017]. In this case, the model breaks the geostrophic balance as fast gravitational waves are superimposed over the large scale waves, and thus the balanced manifold approximation is no longer accurate (i.e it fails to capture the high frequency behavior).

Moreover, standard neural parameterizations (MLPs) have been shown to fail to approximate the closure operator in these dynamics Chekroun et al. [2024]. The authors point out that both the presence of high oscillations and rare events in the attractor introduce a spectral and training bias which prevents regular networks from accurately learning the attractor. This study aims to use a generative neural parameterization to approximate the closure operator and, in this way to provide a surrogate that accurately simulates the multiscale dynamics of the system under this regime.

3 State of the art

- Sanderse et al. [2024] presents a comprehensive review of data-driven techniques for closure problems in multiscale dynamical systems
- Scher and Messori [2019] provide a first glance on the weakness of NNs when extrapolating chaotic high frequency systems. These weaknesses are pointed on Chekroun et al. [2024] which state that MLPs suffer when generalizing regimes dominated by the high-frequencies and extreme events, proposing recurrent or advanced arquitectures in order to capture this behaviour (just like Chattopadhyay et al. [2019])
- Maulik et al. [2019] built a first data-driven closure framework by training a MLP to model local subgrid scale phenomena. Similar frameworks have been proposed in Beck et al. [2019], Kurz and Beck [2020] Park and Choi [2021]. Yang et al. [2019] takes a step further by introducing a PINN to model the small scale perturbation of a LES on a wall-bounded flow
- **Special attention has been recently focused on simulation of extreme events via generative models.** Allouche et al. [2026] show a poor performance of GANs and VAEs when simulating extreme events in financial data. They explain that considering heavy tailed distributions leads to a lack of compactness which violates the assumptions of the Universal approximation theorem. They also propose some improvements (gaussian transformation preprocessing, Tail index parametrization, etc) in order to mitigate this effect.
- Lhaut et al. [2026] develop a GenAI architecture supported on Extreme Values Theory to model heavy tails of finantial data, achieving better results than existing architectures.

4 Methodology

The intuition behind Conditional Flow Matching (CFM) is to learn a velocity field $v_\theta(z, \tau)$ that transforms a simple prior distribution into a target conditional distribution defined by data.

In our case, CFM maps samples from a standard normal prior $p_0 = \mathcal{N}(0, I)$ to the conditional distribution of the fast variables $p_1 = P(X_t, Z_t \mid Y_{hist})$ where $Y_{hist} = [Y_t, Y_{t-\Delta t}, \dots Y_{t-H\Delta t}]$ represents a discrete time historical window of the slow variables. This allows us to transform noise samples into fast variables samples by solving the following Neural ODE along the synthetic time $\tau \in [0, 1]$:

$$\frac{d\phi_\tau}{d\tau} = v_\theta(\phi_\tau, \tau, Y_{hist}), \quad \text{where } \phi_0 \sim \mathcal{N}(0, I) \tag{4}$$

The neural vector field v_θ is trained following the Optimal Transform Conditional Flow Matching (OT-CFM) framework, which seeks minimum energy paths (i.e straight lines) between noise samples $z_0 \sim \mathcal{N}(0, I)$ and data samples $z_1 = [X_t, Z_t]^T \sim P(X_t, Z_t \mid Y_{hist})$. Hence the learned field is given by

$$\phi_\tau = \tau z_1 + (1 - \tau) z_0 \implies \frac{d\phi_\tau}{d\tau} = z_1 - z_0 \tag{5}$$

Therefore, the loss function used for training is the expected squared error between the neural vector field and the OT-CFM target field:

$$J_{CFM}(\theta) = \mathbb{E}_{\tau, z_0, z_1, Y_{hist}}[\|v_\theta(\phi_\tau, \tau, Y_{hist}) - (z_1 - z_0)\|^2] \tag{6}$$

The model for the flow matcher v_θ is a MLP of 3 hidden layers with 256 neurons per layer and SiLU activation function. In the end, we add a linear layer that transforms the features into a flattened vector containing the information of the fast variables X, Z. We consider as baseline data the same dataset described in Liu [2023] (the same used in Chekroun et al. [2024]). The entire dataset contains 6 million points of the fast regime with dt =4.2e-3 model time units (which represent 45 seconds). In order to get a wider landscape of the rare events so the CFM model can learn correctly the dynamics of the fast regime, we take a split of 80% of the simulation dataset for training, and reserve the remaining 20% to perform an out-of-sample *a posteriori* validation.

Algorithm 1 describes the steps for *a posteriori* validation and *online* simulation of the system. Recall that we do not have access to the *true* future trajectory or history of Y. Hence, we can exploit the stochastic nature of CFM to estimate the fast states by initializing an ensemble of trajectories for the slow system. In this way, at each time step k we compute M different guesses for the fast variables by sampling the CFM model conditioned on each trajectory's rolling memory. This gives the samples $X_k^{(m)}$ and $Z_k^{(m)}$ with $m = 1, \dots, M$. Then, for each trajectory, the slow variables $Y_k^{(m)}$ are integrated forward in physical time using an RK4 discretization of the slow equations in (1). In this way we handle the additional uncertainty produced by the high frequencies and the rare events by generating multiple trajectories, which also helps us manage the chaotic nature of the system.

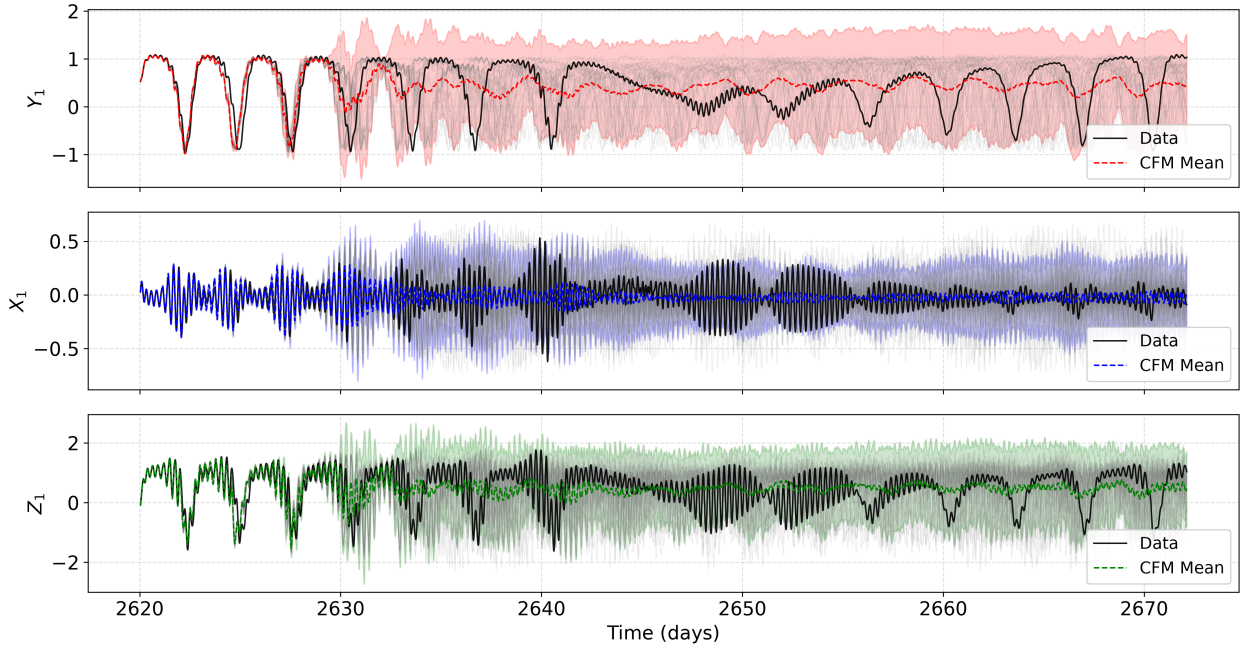
Algorithm 1: A posteriori Validation

Data: Initial history Y_{hist} at $k = 0$, ensemble size M, integration steps N
Result: Ensemble of simulated trajectories for X, Y, Z for $k = 1, \dots, N$
Initialize M ensemble members around Y on $k = 0$;
for $k = 0$ **to** $N - 1$ **do**
  for $m = 1$ **to** M **do**
    Sample noise: $z_0^{(m)} \sim \mathcal{N}(0, I)$;
    Solve Neural ODE: $X_k^{(m)}, Z_k^{(m)} \leftarrow \text{CFM}(z_0^{(m)} \mid Y_{\text{hist}}^{(m)})$
    Advance slow solver: $Y_{k+1}^{(m)} = \text{RK4}(Y_k^{(m)}, X_k^{(m)}, Z_k^{(m)}, \Delta t)$;
  end
  Update memory window: Shift memory $Y_{\text{hist}}^{(m)}$ and set latest state to Y_{k+1};
end

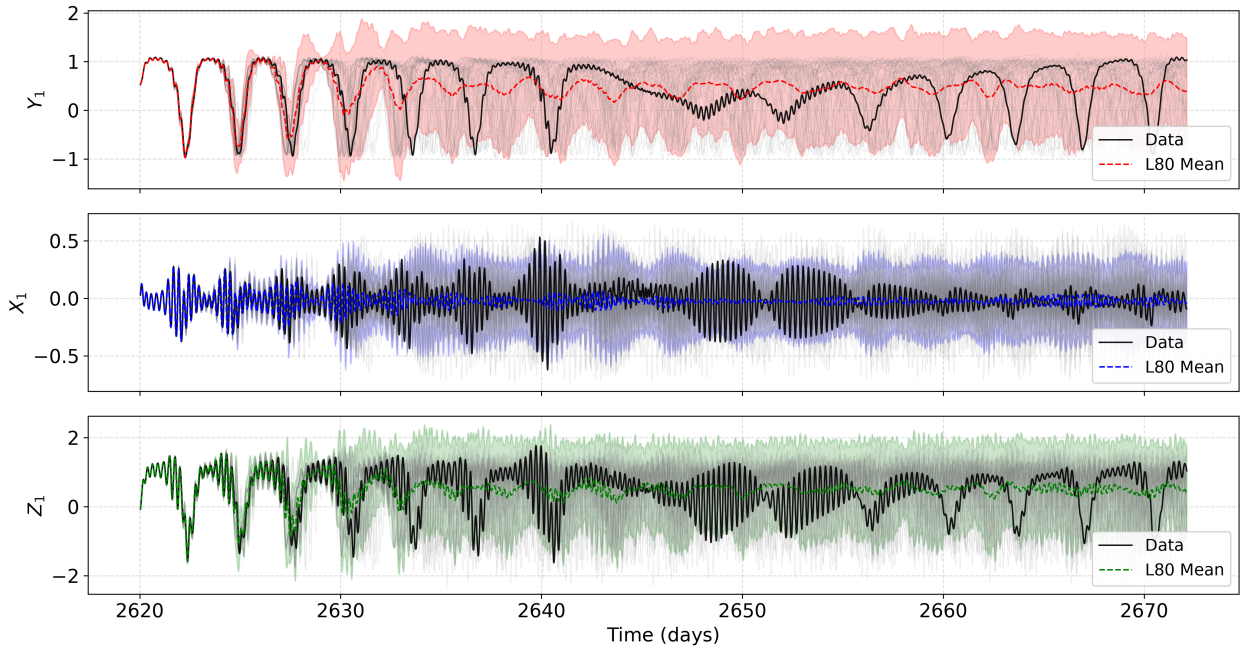
We performed 25 training epochs using the J_{CFM} loss and the Adam optimizer. We choose an ensemble size of $M = 50$ members and memory history Y_{hist} contains information of the previous 5000 points (which is equivalent to $\approx$ 2.6 days) with a stride of 50 to mitigate the computational cost. Thus the actual Y_{hist} contains 100 points.

5 Results

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(a) A posteriori CFM Ensemble Dynamics



(b) L-80 Physical Ensemble Dynamics

Figure 1: A posteriori validation of the Lorenz-80 system. Comparison between the stochastic ensemble generated by the CFM model (upper plot) and the original physical model (lower plot) over 100,000 integration steps.

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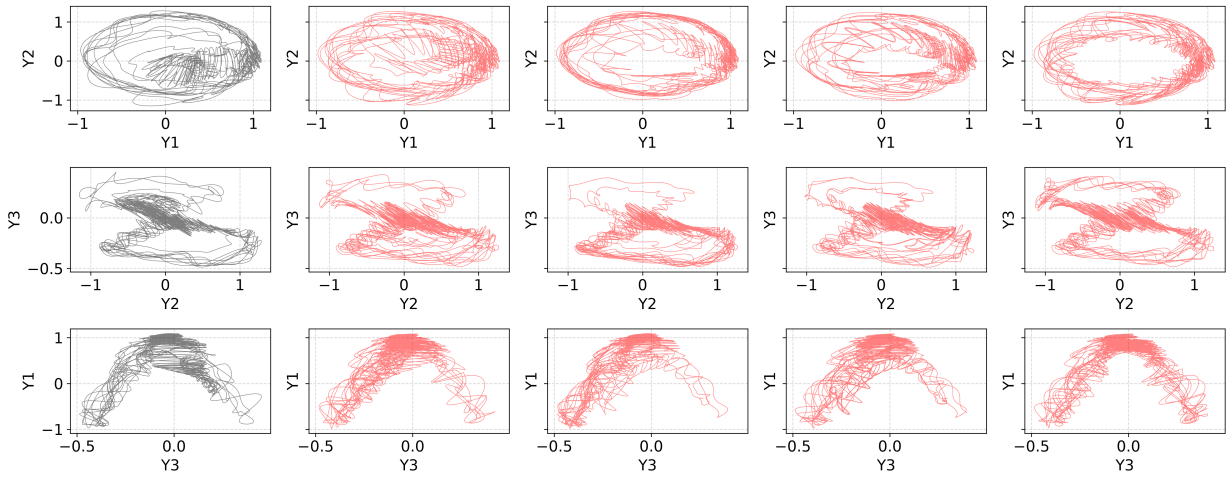


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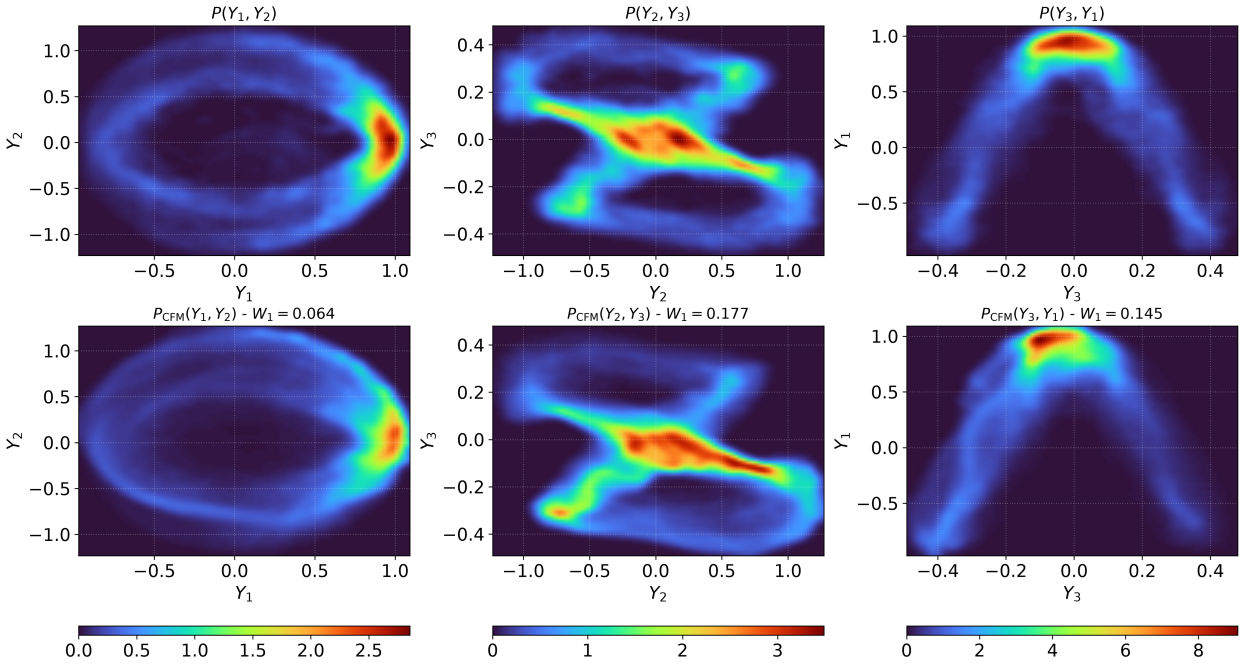


Figure 3: Comparison of the invariant attractor measures of the real and reconstructed dynamics. Here W_1 denotes the 2D Wasserstein distance between the joint distributions of the true (upper row) and CFM-generated variables (lower row)

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6 Conclusions

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